

The `Short` class in Java is a wrapper class for the primitive data type `short`. It provides several utility methods to work with `short` values. This class is part of the `java.lang` package.

Key Features of Short Wrapper Class

1. **Immutability:** Instances of the `Short` class are immutable.
2. **Static Methods:** The `Short` class provides static methods for various operations, such as parsing and comparing shorts.
3. **Short Conversion:** Methods to convert `short` values to other types and vice versa.

Short Class Methods

Here are some commonly used methods in the `Short` class:

1. `short shortValue()`
2. `int compareTo(Short anotherShort)`
3. `static Short decode(String nm)`
4. `boolean equals(Object obj)`
5. `float floatValue()`

6. `int hashCode()`
7. `int intValue()`
8. `long longValue()`
9. `static short parseShort(String s)`
10. `static short parseShort(String s, int radix)`
11. `byte byteValue()`
12. `double doubleValue()`
13. `String toString()`
14. `static String toString(short s)`
15. `static Short valueOf(short s)`
16. `static Short valueOf(String s)`
17. `static Short valueOf(String s, int radix)`

1. short shortValue()

Returns the value of this Short object as a short.

```
public class ShortExample {  
    public static void main(String[]  
args) {  
        Short s = 100;  
        short value = s.shortValue();  
        System.out.println(value);    //
```

Output: 100

```
}  
}
```

Output:

```
100
```

2. `int compareTo(Short anotherShort)`

Compares two `Short` objects numerically.

```
public class ShortExample {  
    public static void main(String[]  
args) {  
        Short s1 = 100;  
        Short s2 = 200;  
  
        System.out.println(s1.compareTo(s2));  
        // Output: -1  
    }  
}
```

Output:

```
-100
```

3. static Short decode(String nm)

Decodes a String into a Short.

```
public class ShortExample {  
    public static void main(String[]  
args) {  
        Short s = Short.decode("100");  
        System.out.println(s);    //  
Output: 100  
    }  
}
```

Output:

100

4. boolean equals(Object obj)

Compares this object to the specified object.

```
public class ShortExample {  
    public static void main(String[]  
args) {  
        Short s1 = 100;  
        Short s2 = 100;
```

```
System.out.println(s1.equals(s2)); //  
Output: true  
    }  
}
```

Output:

```
true
```

5. float floatValue()

Returns the value of this Short object as a float.

```
public class ShortExample {  
    public static void main(String[]  
args) {  
        Short s = 100;  
        float value = s.floatValue();  
        System.out.println(value); //  
Output: 100.0  
    }  
}
```

Output:

```
100.0
```

6. `int hashCode()`

Returns a hash code for this `Short` object.

```
public class ShortExample {  
    public static void main(String[]  
args) {  
        Short s = 100;  
  
        System.out.println(s.hashCode()); //  
Output: 100  
    }  
}
```

Output:

```
100
```

7. `int intValue()`

Returns the value of this `Short` object as an `int`.

```
public class ShortExample {  
    public static void main(String[]  
args) {
```

```
        Short s = 100;
        int value = s.intValue();
        System.out.println(value);    //
Output: 100
    }
}
```

Output:

100

8. long longValue()

Returns the value of this Short object as a long.

```
public class ShortExample {
    public static void main(String[]
args) {
        Short s = 100;
        long value = s.longValue();
        System.out.println(value);    //
Output: 100
    }
}
```

Output:

100

9. static short parseShort(String s)

Parses the string argument as a signed decimal `short`.

```
public class ShortExample {  
    public static void main(String[]  
args) {  
        short s =  
Short.parseShort("100");  
        System.out.println(s);    //  
Output: 100  
    }  
}
```

Output:

100

10. static short parseShort(String s, int radix)

Parses the string argument as a signed `short` in the specified radix.


```
public class ShortExample {
    public static void main(String[]
args) {
        short s =
Short.parseShort("64", 16);
        System.out.println(s);    //
Output: 100
    }
}
```

Output:

100

11. byte byteValue()

Returns the value of this Short object as a byte.

```
public class ShortExample {
    public static void main(String[]
args) {
        Short s = 100;
        byte value = s.byteValue();
        System.out.println(value);    //
Output: 100
    }
}
```

```
}
```

Output:

```
100
```

12. double doubleValue()

Returns the value of this Short object as a double.

```
public class ShortExample {  
    public static void main(String[]  
args) {  
        Short s = 100;  
        double value =  
s.doubleValue();  
        System.out.println(value);    //  
Output: 100.0  
    }  
}
```

Output:

```
100.0
```

13. String toString()

Returns a `String` object representing this `Short`'s value.

```
public class ShortExample {  
    public static void main(String[]  
args) {  
        Short s = 100;  
  
        System.out.println(s.toString()); //  
Output: 100  
    }  
}
```

Output:

```
100
```

14. static String toString(short s)

Returns a `String` object representing the specified `short`.

```
public class ShortExample {  
    public static void main(String[]  
args) {  
        short s = 100;
```

```
System.out.println(Short.toString(s));  
// Output: 100  
    }  
}
```

Output:

100

15. static Short valueOf(short s)

Returns a `Short` instance representing the specified short value.

```
public class ShortExample {  
    public static void main(String[]  
args) {  
        Short s =  
Short.valueOf((short) 100);  
        System.out.println(s);    //  
Output: 100  
    }  
}
```

Output:

100

16. static Short valueOf(String s)

Returns a `Short` object holding the value given by the specified `String`.

```
public class ShortExample {  
    public static void main(String[]  
args) {  
        Short s =  
Short.valueOf("100");  
        System.out.println(s);    //  
Output: 100  
    }  
}
```

Output:

100

17. static Short valueOf(String s, int radix)

Returns a `Short` object holding the value extracted from the specified `String` when parsed with the radix given by the second argument.

```
public class ShortExample {  
    public static void main(String[]  
args) {  
        Short s = Short.valueOf("64",  
16);  
        System.out.println(s);    //  
Output: 100  
    }  
}
```

Output:

100